

Generation Interconnection System Impact Study Report

For Queue Project Q20-1588

Bull Run 138 kV

August 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1588. The Interconnection Customer proposes a 204.6 MW solar photovoltaic facility interconnecting at the Bull Run 138 kV substation in the Virginia area. The request was received on October 06, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 612 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,659,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Cia – Papago Buttes 138.0 kV, the Cia – Papago Buttes 138.0 kV Circuit 1 loads to 122.7% of its 450 MVA rating. Project DFAX: 5.5%.

The Cia – Papago Buttes 138.0 kV Cir. 2 is overloaded to 122.7% of its 450 MVA emergency rating for the loss of the Papago Buttes – Cia Substation 138.0 kV Cir. 1. The project contributes 5.5% DFAX (11.3 MW) to this overload.

Papago Buttes – Mottman 138.0 kV: 121.2% loading (496 MVA) following the outage of the Mottman – Papago Buttes 138.0 kV Line 2; DFAX = 5.8%.

Mottman – Papago Buttes 138.0 kV Circuit 2: 121.2% loading (496 MVA) following the outage of the Mottman – Papago Buttes 138.0 kV; DFAX = 5.8%.

The Idylwood – Falls Church 138.0 kV is overloaded to 112.3% of its 303 MVA emergency rating for the loss of the Centennial – Annandale 138.0 kV Line. The project contributes 10.6% DFAX (21.7 MW) to this overload.

For the loss of Annandale – Centennial 138.0 kV, the Falls Church – Decatur Energy Ctr 138.0 kV line loads to 105.9% of its 247 MVA rating. Project DFAX: 10.6%.

New Hope – Decatur Energy Ctr 138.0 kV: 104.8% loading (219 MVA) following the outage of the Centennial – Annandale 138.0 kV; DFAX = 10.6%.

Holmes Creek – Gallows Rd 138.0 kV Line: 104.2% loading (297 MVA) following the outage of the Idylwood – Gallows Rd 138.0 kV; DFAX = 13.9%.

For the loss of Idylwood – Gallows Rd 138.0 kV L, the Annandale – Holmes Creek 138.0 kV Line loads to 102.5% of its 238 MVA rating. Project DFAX: 13.9%.

For the loss of Darien – Colesville Substation 138.0 kV, the Tokoma – Yulee 138.0 kV loads to 102.4% of its 597 MVA rating. Project DFAX: 5.6%.

The Franconia – Brookridge 138.0 kV is overloaded to 100.3% of its 263 MVA emergency rating for the loss of the Dessau – South Houston 138.0 kV. The project contributes 5.5% DFAX (11.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cia – Papago Buttes 138.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,541,000	31% (\$477,000)
Cia Substation – Papago Buttes 138.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$2,142,000	31% (\$663,000)
Papago Buttes – Mottman 138.0 kV Line 1	Rebuild line with higher-capacity conductor	\$1,153,000	23% (\$265,000)
Mottman – Papago Buttes 138.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,249,000	23% (\$287,000)
Falls Church – Idylwood 138.0 kV	Rebuild line with higher-capacity conductor	\$2,337,000	20% (\$455,000)
Falls Church – Decatur Energy Center 138.0 kV	Reconductor line	\$416,000	20% (\$81,000)
New Hope – Decatur Energy Center 138.0 kV line	Reconductor line	\$195,000	20% (\$38,000)
Holmes Creek – Gallows Rd 138.0 kV Line	Reconductor line	\$610,000	18% (\$113,000)
Annandale – Holmes Creek 138.0 kV Line	Reconductor line	\$713,000	18% (\$132,000)
Tokoma – Yulee 138.0 kV	Reconductor line	\$658,000	15% (\$96,000)
Franconia – Brookridge 138.0 kV	Reconductor line	\$941,000	6% (\$52,000)
	Total allocated to project		\$2,659,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.